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Novel high-throughput screening system for identifying STAT3–SH2 antagonists

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ABSTRACT

Constitutive activation of the oncogenic transcription factor STAT3 frequently occurs in various human malignancies. STAT3 activation involves dimerization via intermolecular pTyr–SH2 interaction. Thus, antagonizing this interaction is a feasible approach to inhibit STAT3 activation for cancer therapy. In order to identify selective STAT3 inhibitors, we developed a biochemical HTS system based on AlphaScreen technology, which measures the abilities of test compounds to antagonize pTyr–SH2 interactions. We screened our chemical libraries using this system and identified 5,15-diphenylporphyrin (5,15-DPP) as a selective STAT3–SH2 antagonist. Selective inhibition of STAT3 nuclear translocation and DNA biding activity was observed in cells treated with 5,15-DPP. IL-6-dependent dimerization of STAT3, c-myc promoter binding and c-myc protein expression were all suppressed by 5,15-DPP, whereas no decrement in either expression or phosphorylation level of STAT3 was observed. Thus, the HTS assay system represented herein may be useful for identifying novel STAT3–SH2 antagonists.

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Signal Transducer and Activator of Transcription (STAT) proteins were originally discovered as latent cytoplasmic transcription factors that mediate cytokine and growth factor-related extracellular signals. These signaling pathways involve the activation of receptor tyrosine kinases, such as epidermal growth factor receptor and platelet-derived growth factor receptor, and Janus kinases (JAKs). Following phosphorylation at tyrosine residue 705 (Tyr705), two STAT monomers dimerize through reciprocal phosphotyrosine and Src homology 2 (SH2) domain interactions. A dimeric form of STAT translocates into the nucleus and binds to STAT-specific DNA-response elements of target genes for transcription activation. The STAT family comprises seven members, namely, STAT1–STAT4, STAT6, and the closely related STAT5a and STAT5b proteins. The members play a role in diverse biological functions, including cell proliferation, cell survival, angiogenesis, apoptosis, and inflammation. Several lines of evidence have implicated some of the STAT family members in malignant transformation and tumor cell survival. In particular, STAT3 is constitutively activated in many types of hematopoietic and solid tumors, such as leukemia and breast and prostate cancers. In light of these findings, STAT3 has been considered an attractive target for the development of new anticancer drugs [1–5]. Because of its central role just downstream of protein tyrosine kinases, aberrant STAT3 activity is often associated with transforming mechanisms induced by oncogenic tyrosine kinases. In addition, recent reports showed that STAT3 was constitutively activated both in tumor cells and in immune cells confined in tumor microenvironments, and that constitutively activated STAT3 inhibited the expression of mediators necessary for immune responses against the tumor cells [6–7]. Although a large number of small molecules have been reported to inhibit STAT3 signaling, the vast majority of them act on STAT3 by inhibiting the upstream tyrosine kinases responsible for STAT3 activation. In particular, small-molecule inhibitors of JAK2 abolished STAT3 activity and induced tumor cell apoptosis [2,3,5,8]. In another study, peptide-based STAT3 inhibitors designed to target the STAT3–SH2 domain were effective in suppressing the cellular functions of STAT3 [3,5,9]. In addition, several small molecules such as STA-21, Stattic, and S3I-201 have been reported to inhibit STAT3 by targeting to the SH2 domain [10–12]. S3I-201 exhibited antitumor efficacy in a mouse xenografted with human breast tumor harboring constitutively active STAT3, and thus presenting a proof-of-concept for the potential clinical use of such small molecules [12].

In order to identify a novel structural class of STAT3 inhibitors, we developed a biochemical assay using an AlphaScreen technology-based HTS system. Using the HTS assay, we identified 5,15-diphenylporphyrin (5,15-DPP) as a selective STAT3–SH2 antagonist. Selective inhibition of STAT3 translocation and DNA biding activity was observed in cells treated with 5,15-DPP. IL-6-dependent dimerization, c-myc promoter binding and c-myc protein expression were all suppressed by 5,15-DPP in a dose-dependent manner, while no decrement in either expression or phosphorylation level of STAT3 was observed. Herein, we report the development of the HTS system and biological activities of 5,15-DPP.

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Materials and methods

Reagents. 5,15-DPP was purchased from Tokyo Chemical Industry (Shinjuku-ku, Tokyo, Japan). Peptides were synthesized by Nippi Biomatrix Laboratory (Adachi-ku, Tokyo, Japan). Anti-FITC acceptor beads and streptavidin-coated donor beads were purchased from PerkinElmer Life Sciences (Waltham, MA, USA). Anti-STAT3, anti-phospho-STAT3 (Tyr705), anti-c-myc, anti-STAT5, and anti-Grb2 antibodies were purchased from Cell Signaling Technologies (Cambridge, MA, USA). Anti-α-tubulin antibody was purchased from Sigma–Aldrich (St. Louis, MO, USA). Horseradish peroxidase (HRP) conjugated anti-mouse IgG, anti-rabbit IgG, and streptavidin–HRP conjugate were obtained from GE Healthcare Bio-Sciences (Piscataway, NJ, USA). *Escherichia coli* XL1-Blue Strain AVB99 was purchased from Avidity (Denver, CO, USA). Plasmid vectors, pAmCyan1-N1 and pZsyellow1-N1, and human universal cDNA library were obtained from Clontech (Mountain View, CA, USA). pBluescript II KS(+) was obtained from Stratagene (La Jolla, CA, USA) and pET-28a (+) was from Novagene (Madison WI, USA). Lipofectamine 2000 was obtained from Invitrogen (Carlsbad, CA, USA).

Cell lines and culture. MDA-MB-468 was purchased from the American Type Culture Collection (Manassas, VA, USA). HEK293 was obtained from the Riken Cell Bank (Tsukuba, Ibaragi, Japan). Cells were maintained in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% FBS, 20 U/ml penicillin, and 20 μg/ml streptomycin at 37 °C in a humidified atmosphere containing 5% CO_2.

Plasmid construction and protein expression. Full-length cDNAs were obtained from a human cDNA library using PCR techniques and cloned into pBluescript II KS(+). The nucleotides coding for human STAT3 in pBluescript II KS(+) was cloned into the HindIII/XhoI sites of modified pET-28a (+). Those of human STAT1 and human Grb2 in pBluescript II (+) were separately cloned into the BamHI/SalI sites of modified pET-28a (+). Avi-tag and 6× His-tag were introduced at the N-terminus of the proteins. *E. coli* XL1-Blue Strain AVB99 transfected with the constructed pET-28a (+) plasmid was grown at 37 °C in LB medium containing 10 μg/ml chloramphenicol and 30 μg/ml kanamycin to an OD_{600} of 0.6. Cells were induced with 0.2 mM isopropyl-β-D-thiogalactopyranoside and 10 μM biotin for 3 h at 30 °C and subsequently harvested by centrifugation. The bacterial pellet was suspended in lysis buffer [50 mM Tris–HCl (pH 7.4), 500 mM NaCl, 5 mM β-mercaptoethanol, 0.1% Nonidet P-40 (NP-40), and 1 Tab/50 ml of complete protease inhibitor cocktail tablets] and lysed by 10 cycles of sonication, each consisting of constant pulse for 1 min at 0 °C. Cell lysates were centrifuged at 12,000*g* for 20 min at 4 °C, and then the supernatant was loaded onto a Ni Sepharose column. Proteins non-specifically bound to the column were washed with washing buffer [50 mM Tris–HCl (pH 7.4), 300 mM NaCl, 0.2 mM EDTA, 0.1% NP-40, 5 mM β-mercaptoethanol, 20 mM imidazole, and 10% glycerol]. Recombinant proteins were eluted with elution buffer [50 mM Tris–HCl (pH 7.4), 300 mM NaCl, 0.2 mM EDTA, 0.1% NP-40, 5 mM β-mercaptoethanol, 10% glycerol, and 10–250 mM imidazole]. Imidazole in the elution fraction was removed by dialysis against 10 mM HEPES (pH 7.4), 125 mM NaCl, 10 mM β-mercaptoethanol, 0.1 mM EGTA, 0.02% NP-40, 0.2 mM PMSF, and 10% glycerol at 4 °C overnight. Purified proteins were analyzed by SDS–PAGE and Western blotting.

AlphaScreen-based assays. AlphaScreen is a bead-based non-radioactive assay system for detecting biomolecular interactions in a microtiter plate format. Binding of biological partners brings donor and acceptor beads into close proximity and as a result, a fluorescent signal between 520 and 620 nm is produced. The AlphaScreen-based assays were performed in a final reaction volume of 25 μl of the assay buffer containing 10 mM HEPES–NaOH (pH 7.4), 50 mM NaCl, 1 mM EDTA (pH 8.0), 0.1% NP-40, and 10 ng/μl BSA in a 96-well microtiter plate at 25 °C. Phospho-Tyr (pTyr) peptide probes used in this study were 5-carboxyfluorescein (FITC)-GpYLPQTV for STAT3, FITC-GpYDKPHVL for STAT1, and FITC-PSpYVNVQN for Grb2. Firstly, 75 nM of each SH2-containing protein was incubated with a test compound for 15 min. Each protein sample was then incubated for 90 min with 50 nM of its corresponding FITC-pTyr peptide, and mixed with streptavidin-coated donor beads and anti-FITC acceptor beads simultaneously before detection at 570 nm using EnVison Xcite (PerkinElmer).

FRET assay. The expression vectors for the fusion proteins STAT3-CFP and STAT3-YFP were prepared by introduction of STAT3 cDNA to pAmCyan1-N1 and pZsyellow1-N1. These vectors were combined at a 1:1 mix ratio and transfected to HEK293 cells with Lipofectamine 2000. Following transfection, the cells were cultured with 2% FBS DMEM for 48 h in a 96-well microtiter plate, treated with 0, 10, 20, or 50 μM 5,15-DPP in a final DMSO concentration of 0.2% for 24 h and then stimulated with 50 ng/ml IL-6 for further 1 h. Fluorescence signals were measured with SpectraMax M5SK (Molecular Devices, Sunnyvale, CA, USA). The signals measured as FRET (excitation at 436 nm, emission at 528 nm) were corrected for crosstalk signals from cyan (excitation at 436 nm, emission at 488 nm) and yellow (excitation at 517 nm, emission at 528 nm).

Treatment of MDA-MB-468 cells with 5,15-DPP. Following overnight serum starvation, MDA-MB-468 cells were treated with 0, 10, 20, or 50 μM 5,15-DPP in a final DMSO concentration of 0.2% for 24 h and then stimulated with 50 ng/ml IL-6 for further 1 h. The cells were washed with ice-cold PBS and were subjected to the following assays.

Quantitative ELISA for transcription factor activation. DNA binding activity of STAT3 was evaluated using a TransAM ELISA kit (Active Motif, Carlsbad, CA, USA) according to the manufacturer's instructions. In brief, 20 μg of the nuclear extract was added to each well of a 96-well microtiter plate coated with an immobilized oligonucleotide containing the STAT3 consensus biding site. One hour later, the wells were incubated with anti-STAT3 antibody for 1 h followed by 1 h of incubation with HRP-conjugated antibody. Thirty minutes after the addition of a developing solution, the reaction was terminated by the addition of H_2SO_4, and absorbance at 450 nm with a reference wavelength of 650 nm was measured using SpectraMax M5SK (Molecular Devices). The nuclear extracts prepared for DNA binding were subjected to Western blotting to analyze the nuclear translocation of endogenous STAT3.

ChIP assay. ChIP assay was performed using an ExactaChIP Human/Mouse STAT3 Chromatin Immunoprecipitation kit (R&D Systems, Minneapolis, MN, USA) according to the manufacturer's protocol. In brief, IL-6-stimulated cells were fixed with 1% formaldehyde and treated with the lysis buffer. After sonication, biotinylated anti-STAT3 antibody or normal sheep IgG was added to the extract followed by precipitation with streptavidin beads. The DNA fraction was eluted from the beads and subjected to PCR with human c-myc primers.

Western blotting. Cells were treated with lysis buffer [20 mM HEPES–NaOH (pH 7.4), 20 mM NaCl, 0.1 mM EDTA (pH 8.0), 5 mM NaF, 10 μM Na_2MoO_4, 0.5% NP-40, and 1 Tab/50 ml complete protease inhibitor cocktail tablets]. Forty micrograms of total proteins was subjected to SDS–PAGE and then transferred onto a nitrocellulose membrane. Membranes were blocked in 5% skim milk in TBS/0.1% Tween 20. They were then incubated with primary antibodies diluted at 1:1000 in 1.5% skim milk in TBS (pH 7.4)/0.1% Tween 20 overnight at 4 °C, followed by incubation with HRP-conjugated secondary antibodies diluted at 1:5000 in TBS (pH 7.4)/0.1% Tween 20 for 1 h at room temperature. Immunoblots were developed using Chemiluminescent Substrate.

Results and discussion

Development of a HTS system for STAT3 inhibitors

Since previous studies showed that AlphaScreen technology was capable of analyzing protein–protein or protein–peptide interactions [13–14], we hypothesized that the AlphaScreen system could be used for detecting the interaction between the SH2 domain and the pTyr-containing peptide. In order to develop the assay, SH2-containing recombinant proteins STAT3, STAT1, and Grb2 were expressed and subsequently biotinylated through avi-tag [15] introduced at the N-terminus of the proteins. Evaluation by immunoblotting using antibodies specific for STAT3, STAT1, and Grb2 clearly showed successful expression and biotinylation of the recombinant proteins (Fig. 1A). The sequences of the pTyr peptides as SH2 ligands used in this study were the same as those used in previous studies [16–18]. When 75 nM biotinylated STAT3 and 50 nM FITC-GpYLPQTV were combined in the assay mixture, the maximum chemiluminescence signal was generated from the reaction of the streptavidin-coated donor beads and anti-FITC acceptor beads. A non-labeled inhibitor peptide, Ac-GpYLPQTV-NH_2 [16], competitively and strongly inhibited the binding of STAT3 to FITC-GpYLPQTV (IC_{50} = 0.3 μM), whereas its non-phospho counterpart, Ac-GYLPQTV-NH_2, or any other peptides used in this study did not (Fig. 1B and Table 1). STAT1 and Grb2 with their corresponding peptides showed similar results and no crossover inhibition among those phosphopeptides was observed (Table 1). These results show that the chemiluminescence signal detected in the assay indicates specific interaction between the SH2-containing protein and its target peptide, and thus suggest that this system is capable of screening test compounds for the discovery of SH2 antagonists.

Table 1
IC_{50} values of STAT3, STAT1, and Grb2 inhibition by 5,15-DPP and peptides as determined by AlphaScreen assays.

Compounds	IC_{50} (μM)		
	STAT3	STAT1	Grb2
5,15-DPP	0.28	10	>20
Ac-GpYLPQTV-NH_2	0.33	>20	>20
Ac-GYLPQTV-NH_2	>20	>20	>20
Ac-GpYDKPHVL-NH_2	>20	1.6	>20
Ac-GYDKPHVL-NH_2	>20	>20	>20
Ac-PSpYVNVQN-NH_2	>20	>20	1.9
Ac-PSYVNVQN-NH_2	>20	>20	>20

AlphaScreen assays were performed as described in Materials and methods, and the IC_{50} values were determined from dose–response curves (n = 3).

Identification of 5,15-DPP as an antagonist of STAT3–SH2

We then screened our chemical library (containing more than 10,000 small molecules) using the *in vitro* AlphaScreen assay and identified several hit compounds that had the ability to inhibit the STAT3 binding to phosphopeptides in a concentration-dependent manner. One of them is a compound structurally related to porphyrin, 5,15-DPP (Fig. 1C). This compound selectively antagonized STAT3–SH2 (IC_{50} = 0.28 μM) *in vitro* over the other SH2-containing proteins STAT1 and Grb2 (Fig. 1D and Table 1). Reduced

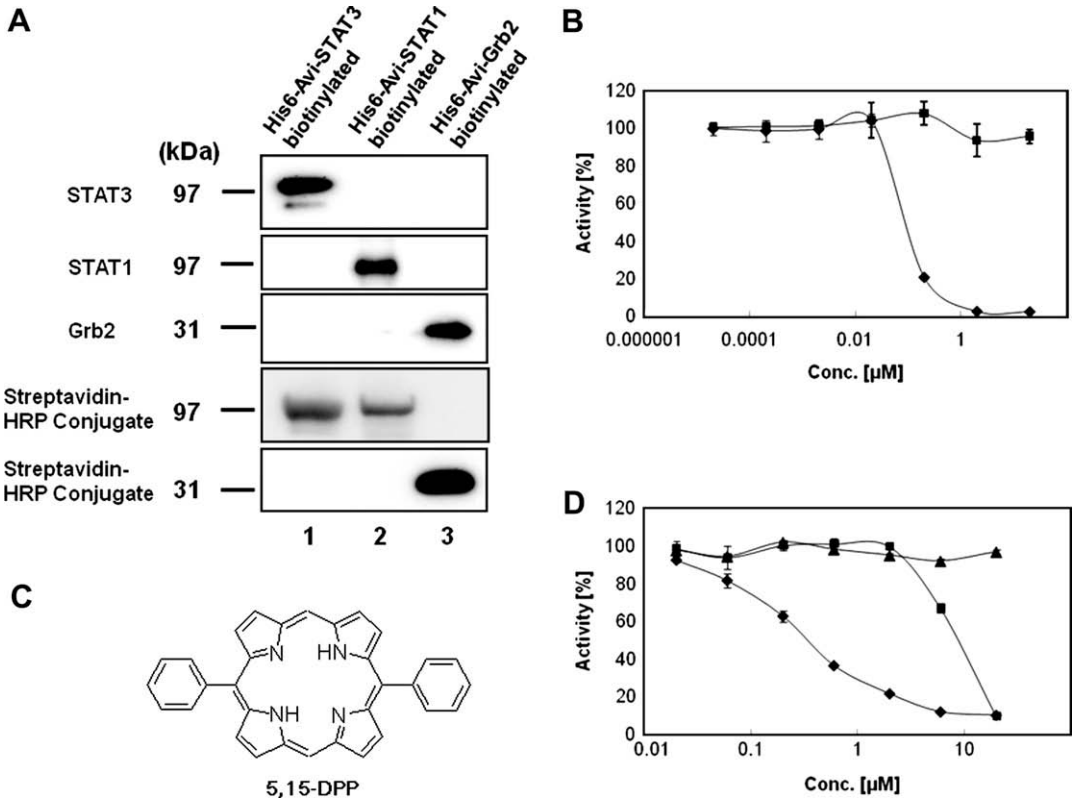


Fig. 1. Development of assay system and identification of 5,15-DPP. (A) Western blotting analysis of purified biotinylated proteins. His6-Avi-tagged STAT3 (lane 1), His6-Avi-tagged STAT1 (lane 2) and His6-Avi-tagged Grb2 (lane 3) were resolved on SDS–PAGE, then probed with antibodies as indicated and streptavidin–HRP conjugate. (B) Concentration- and phosphorylation-dependent inhibition of interaction between STAT3 and the FITC-GpYLPQTV by Ac-GpYLQTV-NH_2 (♦) and Ac-GYLQTV-NH_2 (■). (C) Chemical structure of 5,15-DPP. (D) Inhibitory activity of 5,15-DPP against SH2-containing proteins STAT3 (♦), STAT1 (■), and Grb2 (▲) in the AlphaScreen-based assay. Competitive AlphaScreen-based assays (B and D) were carried out as described in Materials and methods. The graphs were generated from the means ± SD of three separate experiments.

inhibition of SH2 function in STAT1 and Grb2 induced by 5,15-DPP correlated with the degree of sequence similarity (78% and 65%, respectively) in the SH2 domain among the three proteins. Another biochemical analysis with Surface Plasmon Resonance (SPR) determined the direct binding of 5,15-DPP to the recombinant STAT3 (data not shown). From the kinetic analyses using SPR, the estimated K_D values for the STAT3 binding to the Ac-GpYLPQTV-NH_2 and 5,15-DPP were 19 nM and 880 nM, respectively. Although the affinity of 5,15-DPP to STAT3 was lower than that of the phosphopeptide ligand, these results suggest that 5,15-DPP directly binds to STAT3 and antagonizes the function of STAT3–SH2 *in vitro*.

Inhibition of STAT3 activity by 5,15-DPP in cells

In order to analyze the ability of 5,15-DPP to inhibit STAT3 dimerization in cells, a fluorescence resonance energy transfer (FRET) assay was performed. HEK293 cells were co-transfected with both STAT3-CFP and STAT3-YFP. IL-6 stimulation induced a high FRET signal by the interaction between STAT3-CFP and STAT3-YFP in the cells. Pretreatment of the cells with 5,15-DPP before IL-6 stimulation resulted in the reduction of FRET signals, as shown in Fig. 3A.

We next examined whether 5,15-DPP inhibited STAT3 nuclear translocation and subsequent DNA binding in human breast cancer cells with activated STAT3. An ELISA-based assay with the consensus-binding site on the oligonucleotide immobilized in a 96-well microtiter plate was used to analyze the nuclear lysates from MDA-MB-468 cells treated with 0 μM, 10 μM, 20 μM or 50 μM 5,15-DPP. The DNA binding activity of STAT3 was suppressed by the treatment with 5,15-DPP in a concentration-dependent manner, as shown in upper panel in Fig. 2A. STAT3 nuclear translocation was also inhibited by 5,15-DPP (Fig 2A, lower panel), suggesting that inhibition of STAT3 activation blocks its nuclear localization. In contrast, no inhibition was observed in the nuclear translocation and DNA binding of STAT1, STAT5a, or STAT5b (Fig. 2B–D).

In order to elucidate the effect of 5,15-DPP on the STAT3–DNA complex in cells, a chromatin immunoprecipitation (ChIP) assay was conducted. Following stimulation of cells with IL-6 for 1 h, protein–DNA complexes were fixed and immunoprecipitated with anti-STAT3 antibody. The complex was subjected to PCR amplification for the detection of the c-myc promoter, which is known as one of the targets for transcriptional activation of STAT3. The amplified c-myc promoter was clearly detected in the absence of 5,15-DPP, whereas a reduction in the band intensity was observed in the presence of this compound (Fig. 3B). These results support the hypothesis that 5,15-DPP inhibits STAT3 dimerization by acting as an antagonist at the SH2 domain and thus blocks its DNA binding events that is required for the transcriptional activity of STAT3 in cells. The expression level of c-myc protein was also decreased in the 5,15-DPP-treated cells (Fig. 3C). However, the phosphorylation of STAT3 was not affected. Taken together, the inhibition of STAT3 function by 5,15-DPP in MDA-MB-468 cells may be caused by the direct inhibition of STAT3 dimerization. Further analysis is necessary to elucidate the exact mode of action of 5,15-DPP in the cells.

In conclusion, we developed a HTS system based on AlphaScreen technology in order to identify a novel structural class of antagonists of STAT3–SH2 function. Using this system, we

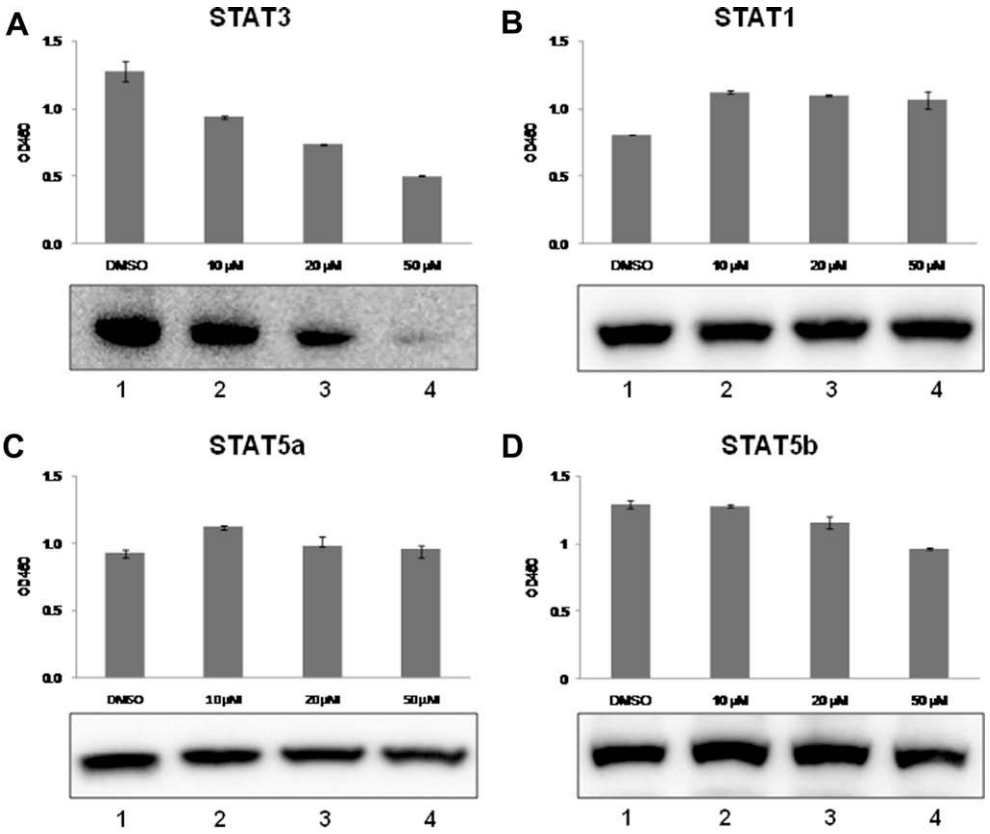


Fig. 2. Effect of 5,15-DPP on nuclear translocation and DNA biding activities of STAT3 (A), STAT1 (B), STAT5a (C), and STAT5b (D). Serum-starved MDA-MB-468 cells were treated with the indicated concentration of 5,15-DPP for 24 h and then stimulated with 50 ng/ml IL-6 for 1 h. The nuclear extract was prepared to analyze the DNA binding (upper panel) and the nuclear translocation (lower panel) of STATs. DNA binding activity assays were performed as described in Materials and methods. The graphs were generated from the means ± SD of three separate experiments.

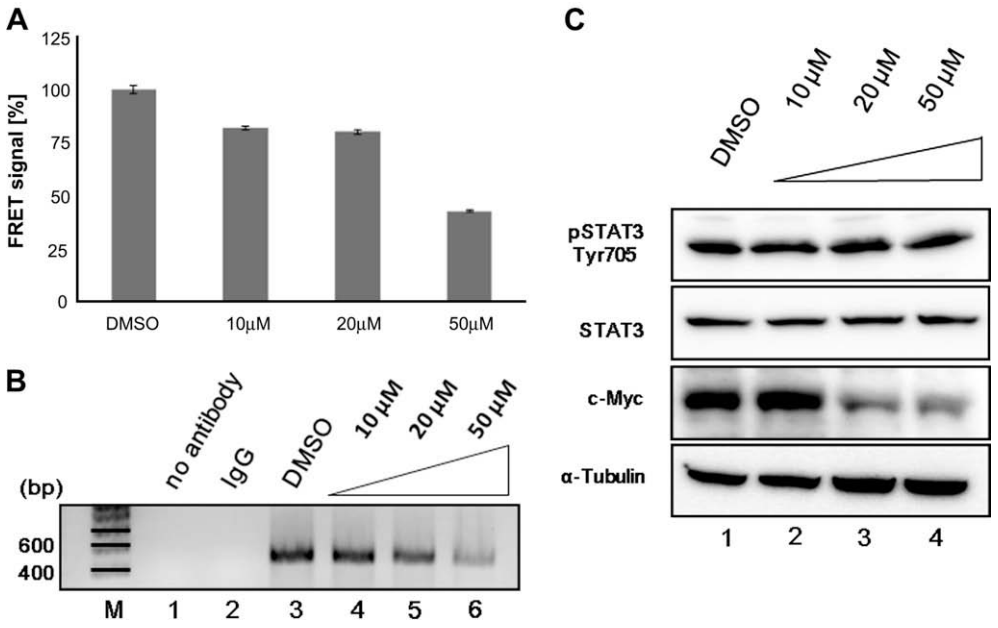


Fig. 3. Inhibition of STAT3 function by 5,15-DPP in STAT3-activated human cancer cells. (A) Effect of 5,15-DPP on the FRET signals generated from interactions between STAT3-CFP and STAT3-YFP in HEK293 cells. The transient transfectant was treated with the indicated concentrations of 5,15-DPP for 24 h. Following stimulation by IL-6 for 1 h, FRET signals were measured as described in Materials and methods. (B) Effect of 5,15-DPP on the c-myc promoter binding by STAT3 in MDA-MB-468 cells. MDA-MB-468 cells were exposed to the indicated concentrations of 5,15-DPP for 24 h before IL-6 stimulation. Following stimulation, the cells were subjected to ChIP assay. Normal sheep IgG was used as a negative control. (C) Effect of 5,15-DPP on Tyr705 phosphorylation of STAT3, and expressions of STAT3 and c-myc protein. MDA-MB-468 cells were treated with the indicated concentrations of 5,15-DPP for 24 h and lysed for Western blotting analysis.

screened our chemical libraries and identified 5,15-DPP. 5,15-DPP inhibited the cellular function of STAT3 without inhibiting upstream JAK kinase activity, presumably due to STAT3 dimerization inhibition by blocking the SH2 domain. Thus, our HTS system may be useful for identifying novel STAT3 inhibitors.

Acknowledgments

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